WoSM Academy

MAT231-F27 · CALCULUS II

The chain rule

The multivariable chain rule and the Jacobian matrix

Dr. Ali Khudiyev
WoSM Academy

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Last time

- The gradient ∇f: $D_{\mathbf{v}} f = \nabla f \cdot \mathbf{v}$; it points in the direction of steepest ascent.
- ∇f is perpendicular to level sets; tangent lines and planes.
- Vector rules such as $\nabla(\mathbf{x}^\top A\mathbf{x}) = (A + A^\top)\mathbf{x}$; least squares.

Today: the most important rule of multivariable calculus. Most functions are built by composing simpler ones — and so are their derivatives. The chain rule is also the mathematics behind backpropagation in neural networks.

This lecture

01 The Jacobian matrix

02 The chain rule

03 Special cases

04 Implicit differentiation revisited

05 Computational graphs and backpropagation

06 The Jacobian determinant

07 Summary and practice

Reading: G. Strang, E. Herman, *Calculus, Volume 3* (OpenStax), Section 4.5.

01

The Jacobian matrix

The Jacobian matrix

For $F = (F_1, \dots, F_m) \colon U \subseteq \mathbb{R}^n \to \mathbb{R}^m$ differentiable at $\mathbf{a}$, the derivative $DF(\mathbf{a})$ is the $m \times n$ **Jacobian matrix:**

$$DF(\mathbf{a}) = \begin{pmatrix} \nabla F_1(\mathbf{a})^\top \\ \vdots \\ \nabla F_m(\mathbf{a})^\top \end{pmatrix} = \begin{pmatrix} \frac{\partial F_1}{\partial x_1} & \cdots & \frac{\partial F_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial F_m}{\partial x_1} & \cdots & \frac{\partial F_m}{\partial x_n} \end{pmatrix}(\mathbf{a}).$$

Row i is the gradient of the component F_i; column j says how F moves when only x_j moves.

- $m = 1$: $Df = \nabla f^\top$, a row. $n = 1$, a curve $\mathbf{r}(t)$: $D\mathbf{r} = \mathbf{r}'(t)$, a column.
- A linear map $F(\mathbf{x}) = M\mathbf{x}$ has $DF(\mathbf{a}) = M$ at every point.

Examples

Polar coordinates, $\mathbb{R}^2 \to \mathbb{R}^2$:
$P(r, \theta) = (r\cos\theta,\ r\sin\theta)$,

$$DP(r,\theta) = \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}.$$

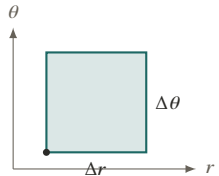
$\mathbb{R}^2 \to \mathbb{R}^3$:
$F(x, y) = (x^2 y,\ x + y^3,\ e^x)$,

$$DF(x,y) = \begin{pmatrix} 2xy & x^2 \\ 1 & 3y^2 \\ e^x & 0 \end{pmatrix}.$$

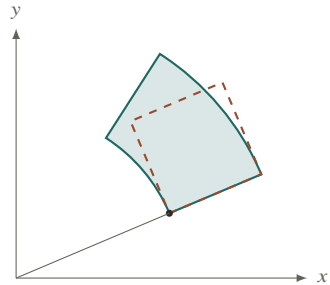
The best linear approximation

$F(\mathbf{a}+\mathbf{h}) = F(\mathbf{a}) + DF(\mathbf{a})\,\mathbf{h} + o(\|\mathbf{h}\|)$: near $\mathbf{a}$, F acts like the linear map $DF(\mathbf{a})$ (Linear Algebra, Units 9–10).

A picture: polar coordinates near a point



(r, θ)-plane: $\Delta r \times \Delta\theta$



(x, y)-plane: the image, and DP applied to the rectangle

The image of a small rectangle is a curved patch. The Jacobian maps the rectangle to a parallelogram that matches the patch to first order — better and better as the rectangle shrinks.

02

The chain rule

The chain rule

Theorem (Chain rule)

Let $G\colon \mathbb{R}^k \to \mathbb{R}^n$ be differentiable at $\mathbf{a}$, and $F\colon \mathbb{R}^n \to \mathbb{R}^m$ differentiable at $G(\mathbf{a})$. Then $F \circ G$ is differentiable at $\mathbf{a}$, and $D(F \circ G)(\mathbf{a}) = DF\big(G(\mathbf{a})\big)\, DG(\mathbf{a})$.

- The derivative of a composition is the *product* of the Jacobian matrices; the sizes $(m \times n)(n \times k) = m \times k$ always fit.
- One variable: $(f \circ g)'(a) = f'(g(a))\, g'(a)$ (Calculus I, Unit 10).

Why it must be so

Near $\mathbf{a}$, G acts like the matrix $DG(\mathbf{a})$, and near $G(\mathbf{a})$, F like $DF(G(\mathbf{a}))$. Composing linear maps multiplies their matrices (Linear Algebra, Unit 3).

Proof

Proof.

Let $\mathbf{b} = G(\mathbf{a})$, $A = DG(\mathbf{a})$, $B = DF(\mathbf{b})$, and write $G(\mathbf{a}+\mathbf{h}) = \mathbf{b} + A\mathbf{h} + \rho(\mathbf{h})$ and $F(\mathbf{b}+\mathbf{k}) = F(\mathbf{b}) + B\mathbf{k} + \sigma(\mathbf{k})$, with $\rho(\mathbf{h}) = o(\|\mathbf{h}\|)$ and $\sigma(\mathbf{k}) = o(\|\mathbf{k}\|)$. Put $\mathbf{k} = G(\mathbf{a}+\mathbf{h}) - \mathbf{b} = A\mathbf{h} + \rho(\mathbf{h})$:

$$F(G(\mathbf{a}+\mathbf{h})) = F(\mathbf{b}) + BA\,\mathbf{h} + \underbrace{B\rho(\mathbf{h})}_{(1)} + \underbrace{\sigma(\mathbf{k})}_{(2)}.$$

(1) $\|B\rho(\mathbf{h})\| \le C\|\rho(\mathbf{h})\| = o(\|\mathbf{h}\|)$, since a matrix stretches vectors by at most a constant C. (2) For small $\mathbf{h}$, $\|\mathbf{k}\| \le C'\|\mathbf{h}\|$, so $\frac{\|\sigma(\mathbf{k})\|}{\|\mathbf{h}\|} \le C' \frac{\|\sigma(\mathbf{k})\|}{\|\mathbf{k}\|} \to 0$ (and $\sigma(\mathbf{0}) = \mathbf{0}$). So the remainder is $o(\|\mathbf{h}\|)$, and $D(F \circ G)(\mathbf{a}) = BA$. □

03

Special cases

A function along a curve

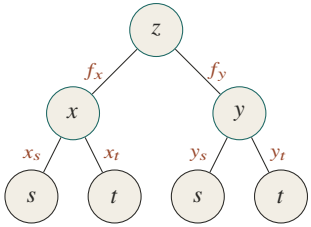
$f\colon \mathbb{R}^2 \to \mathbb{R}$ along a curve $\mathbf{r}(t) = (x(t), y(t))$ — say a temperature along your path. With $(1 \times 2)(2 \times 1) = 1 \times 1$:

$$\frac{\mathrm{d}}{\mathrm{d}t} f(x(t), y(t)) = f_x\, x'(t) + f_y\, y'(t) = \nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t).$$

Example

$f(x, y) = x^2 y$ on the circle $\mathbf{r}(t) = (\cos t, \sin t)$: $\frac{\mathrm{d}}{\mathrm{d}t} f(\mathbf{r}(t)) = 2xy \cdot (-\sin t) + x^2 \cdot \cos t = -2\cos t \sin^2 t + \cos^3 t$. Check: $f(\mathbf{r}(t)) = \cos^2 t \sin t$ has the same derivative.

Two intermediate variables



A dependency tree

$z = f(x, y)$ with $x = x(s, t)$, $y = y(s, t)$:

$$z_s = f_x\, x_s + f_y\, y_s,$$
$$z_t = f_x\, x_t + f_y\, y_t.$$

- Rule: for z_s, follow every path from z down to s, multiply along each path, and add the paths.
- In matrices: $(z_s \;\; z_t) = (f_x \;\; f_y) \begin{pmatrix} x_s & x_t \\ y_s & y_t \end{pmatrix}$.

Example: partial derivatives in polar coordinates

With $x = r\cos\theta$, $y = r\sin\theta$, a function $f(x, y)$ becomes a function of (r, θ):

$$f_r = f_x\cos\theta + f_y\sin\theta, \qquad f_\theta = -f_x\, r\sin\theta + f_y\, r\cos\theta.$$

Example

$f = x^2 + y^2$: $f_r = 2x\cos\theta + 2y\sin\theta = 2r$. Indeed $f = r^2$.

Inverting the Jacobian of P (possible for $r > 0$) gives the other direction:

$$f_x = \cos\theta\, f_r - \frac{\sin\theta}{r} f_\theta, \qquad f_y = \sin\theta\, f_r + \frac{\cos\theta}{r} f_\theta.$$

For example, the angle θ itself has $\theta_x = -\frac{\sin\theta}{r} = -\frac{y}{x^2+y^2}$.

04

Implicit differentiation revisited

Implicit differentiation

Suppose the curve $F(x, y) = 0$ is the graph of a differentiable function $y = y(x)$ near a point. Differentiate $F(x, y(x)) = 0$ with the chain rule:

$$F_x + F_y\, y'(x) = 0 \qquad \Longrightarrow \qquad y'(x) = -\frac{F_x}{F_y} \quad (F_y \neq 0).$$

Example

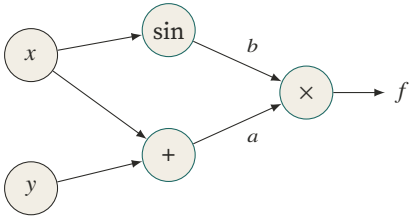
- $x^2 + y^2 = 25$ at $(3, 4)$: $y' = -\frac{2x}{2y} = -\frac{3}{4}$ — as in Calculus I.
- $x^3 + y^3 = 6xy$ at $(3, 3)$: $F_x = 3x^2 - 6y = 9$, $F_y = 3y^2 - 6x = 9$, so $y' = -1$.

The tangent direction $(F_y, -F_x)$ is perpendicular to ∇F (Lecture 11). We *assumed* that $y(x)$ exists; the implicit function theorem (Lecture 17) proves it when $F_y \neq 0$.

05

Computational graphs and backpropagation

A computational graph



$f(x, y) = (x + y)\sin x$ as elementary steps:

$$a = x + y, \qquad b = \sin x, \qquad f = a \cdot b.$$

Each node knows its own *local* derivatives:

$$\frac{\partial a}{\partial x} = 1, \quad \frac{\partial a}{\partial y} = 1, \quad \frac{\mathrm{d}b}{\mathrm{d}x} = \cos x,$$
$$\frac{\partial f}{\partial a} = b, \quad \frac{\partial f}{\partial b} = a.$$

The chain rule multiplies local derivatives along paths and adds over paths. In which order?

Forward mode

Pick one input and push derivatives forward with the values. For $\frac{\partial}{\partial x}$, start with $\dot{x} = 1$, $\dot{y} = 0$:

$$\dot{a} = \dot{x} + \dot{y} = 1, \qquad \dot{b} = \cos x \cdot \dot{x} = \cos x, \qquad \dot{f} = b\,\dot{a} + a\,\dot{b} = \sin x + (x + y)\cos x.$$

> **Example (At $(x, y) = (1, 2)$)**
>
> Values: $a = 3$, $b = \sin 1 \approx 0.8415$, $f \approx 2.5244$. Then $f_x = 0.8415 + 3 \cdot 0.5403 \approx 2.4624$. For f_y, a second pass with $\dot{x} = 0$, $\dot{y} = 1$: $f_y = \sin 1 \approx 0.8415$.

One pass per input variable. Fine for two inputs; hopeless for a million.

Reverse mode: backpropagation

First compute all values forward. Then push **adjoints** $\bar{u} = \frac{\partial f}{\partial u}$ backwards, from the output to the inputs, starting with $\bar{f} = 1$:

$$\bar{a} = \bar{f} \cdot b = \sin x, \qquad \bar{b} = \bar{f} \cdot a = x + y,$$

$$\bar{x} = \bar{a} \cdot 1 + \bar{b} \cdot \cos x = \sin x + (x + y)\cos x, \qquad \bar{y} = \bar{a} \cdot 1 = \sin x.$$

- *One* backward pass gives *all* partial derivatives: $\bar{x} = f_x$, $\bar{y} = f_y$.
- Where a variable feeds several nodes (x feeds a and b), its adjoint is the *sum* over the paths — the "add over paths" of the chain rule.
- At $(1, 2)$: $\bar{x} \approx 2.4624$, $\bar{y} \approx 0.8415$, as before.

Why reverse mode trains neural networks

A network computes a loss $L\colon \mathbb{R}^n \to \mathbb{R}$ through layers, $L = \ell \circ F_k \circ \cdots \circ F_1$, with n often in the millions. By the chain rule,

$$DL = D\ell \cdot DF_k \cdots DF_1 \qquad (1 \times n).$$

- **Forward mode** multiplies from the right: one pass for each of the n input directions.
- **Reverse mode** multiplies from the left, starting with the row $D\ell$: every step is a row vector times a matrix. One backward pass gives the whole gradient.
- So the full gradient costs a small constant times one evaluation of L, however large n is. Backpropagation is exactly this.

06

The Jacobian determinant

The Jacobian determinant

For $F\colon \mathbb{R}^n \to \mathbb{R}^n$ the Jacobian matrix is square, and $\det DF(\mathbf{a})$ is the **Jacobian determinant**.

- A linear map M multiplies volumes by $|\det M|$ (Linear Algebra, Unit 12).
- Near $\mathbf{a}$, F acts like $DF(\mathbf{a})$. So a small region of volume V near $\mathbf{a}$ is mapped to a region of volume about $|\det DF(\mathbf{a})|\,V$: the local scaling factor of volumes.
- The sign tells whether F preserves or reverses orientation.
- By the chain rule and $\det(BA) = \det B \det A$, the factors multiply under composition.

Polar coordinates, again

$$\det DP(r,\theta) = \det \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix} = r\cos^2\theta + r\sin^2\theta = r.$$

- The patch in the picture ($1 \le r \le 1.6$, $0.4 \le \theta \le 1$) has area $\frac{1}{2}(1.6^2 - 1^2)\cdot 0.6 = 0.468$, and $r\,\Delta r\,\Delta\theta$ at $r = 1.3$ is $1.3 \cdot 0.6 \cdot 0.6 = 0.468$.
- Patches near the origin are squeezed ($r \to 0$); far away they are stretched.

Coming up

Change of variables in integrals (Lecture 20): $\iint f(x,y)\,\mathrm{d}x\,\mathrm{d}y = \iint f(r\cos\theta, r\sin\theta)\,r\,\mathrm{d}r\,\mathrm{d}\theta$. The inverse function theorem (Lecture 17): if $\det DF(\mathbf{a}) \neq 0$, F is invertible near $\mathbf{a}$.

07

Summary and practice

Summary

- The **Jacobian matrix** $DF(\mathbf{a})$: rows are the gradients of the components; it is the matrix of the best linear approximation.
- **Chain rule**: $D(F \circ G)(\mathbf{a}) = DF(G(\mathbf{a}))\, DG(\mathbf{a})$ — composing linear approximations multiplies matrices.
- Along a curve: $\frac{\mathrm{d}}{\mathrm{d}t} f(\mathbf{r}(t)) = \nabla f \cdot \mathbf{r}'$. Through intermediate variables: multiply along paths, add over paths.
- Implicit differentiation: $y' = -F_x / F_y$.
- Computational graphs: forward mode pushes derivatives with the values; reverse mode (backpropagation) pushes adjoints back and gets the whole gradient in one pass.
- $|\det DF|$ is the local factor by which F scales volumes; for polar coordinates it is r.

Practice

1. Find the Jacobian matrix and its determinant for $F(x, y, z) = (xy,\ yz,\ zx)$.
2. Let $z = e^{xy}$ with $x = s + t$, $y = s - t$. Find z_s and z_t with the chain rule, and check by substituting.
3. Find $\frac{\mathrm{d}}{\mathrm{d}t} f(\mathbf{r}(t))$ for $f(x, y) = xy^2$ and $\mathbf{r}(t) = (e^t, t^2)$.
4. Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ on the curve $x^2 y + y^3 = 2$ at $(1, 1)$.
5. Show that if $f(x, y) = g(x^2 + y^2)$ for a differentiable g, then $y f_x - x f_y = 0$.
6. Compute ∇f for $f(x, y) = xy + e^x$ at $(0, 3)$ by forward mode and by reverse mode.
7. Show that the Jacobian determinant of spherical coordinates $(\rho, \varphi, \theta) \mapsto (\rho \sin\varphi \cos\theta,\ \rho \sin\varphi \sin\theta,\ \rho \cos\varphi)$ is $\rho^2 \sin\varphi$.
8. If F is invertible with a differentiable inverse, show that $D(F^{-1})(F(\mathbf{a})) = DF(\mathbf{a})^{-1}$.

Answers

1. $DF = \begin{pmatrix} y & x & 0 \\ 0 & z & y \\ z & 0 & x \end{pmatrix}$, $\det DF = 2xyz$.
2. $z_s = ye^{xy} + xe^{xy} = 2s\,e^{s^2-t^2}$, $z_t = (y-x)e^{xy} = -2t\,e^{s^2-t^2}$; indeed $z = e^{s^2-t^2}$.
3. $y^2x' + 2xy\,y' = t^4e^t + 4t^3e^t$.
4. $F_x = 2xy = 2$, $F_y = x^2 + 3y^2 = 4$: $y' = -\tfrac{1}{2}$.
5. $f_x = 2x\,g'(x^2+y^2)$ and $f_y = 2y\,g'(x^2+y^2)$.
6. Graph $a = xy$, $b = e^x$, $f = a + b$. Both modes give $f_x = y + e^x = 4$, $f_y = x = 0$.
7. Expand the 3×3 determinant along its last row $(\cos\varphi,\ -\rho\sin\varphi,\ 0)$ and use $\sin^2 + \cos^2 = 1$.
8. $F^{-1} \circ F$ is the identity, so the chain rule gives $D(F^{-1})(F(\mathbf{a}))\,DF(\mathbf{a}) = I$.

Next time

Lecture 13 — Second derivatives (Unit 13)

- Higher partial derivatives; Schwarz's theorem
- The Hessian matrix
- The Taylor expansion in several variables; quadratic forms and curvature

Reading: J. E. Marsden, A. Tromba, *Vector Calculus*, Sections 3.1 and 3.2.

Questions: the module channel on Discord, or office hours, Monday–Friday, 10:00–17:00 (Baku time, UTC+4), ali.khudiyev@wosm.academy.